LAB Assignment No 4

Interfacing LED with ESP32 and Web Dashboard Control

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Objective:

- Learn how to control an LED connected to ESP32 using a web-based dashboard.
- Understand communication between ESP32 and Firebase Realtime Database.
- Observe real-time LED control via web interface**.**

Materials Required:

1. ESP32 (simulated in Wokwi)
2. LED
3. 220Ω resistor
4. Breadboard and jumper wires
5. Computer with internet access (for Wokwi simulation and Firebase)
6. Web browser

Lab Procedure

Step 1: Setup Wokwi ESP32 Simulation

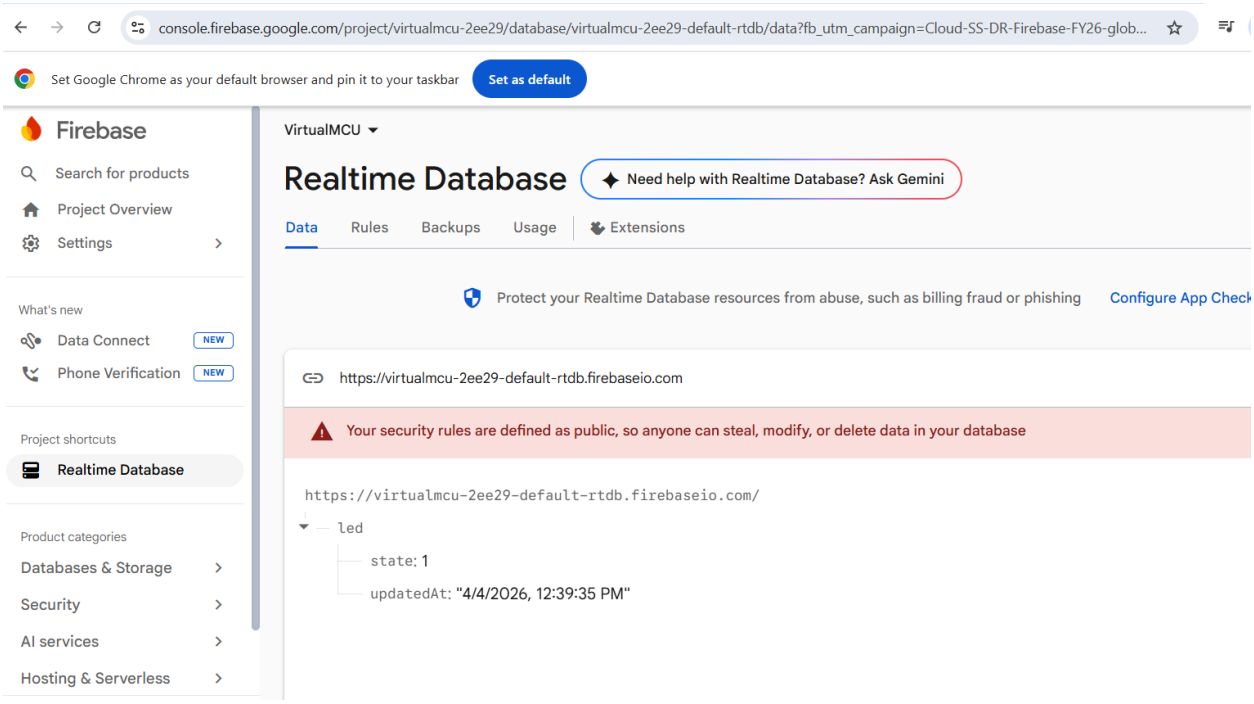
1. Open Wokwi online simulator.
2. Add an ESP32 module.
3. Connect an LED to **GPIO 2** of ESP32 using a 220Ω resistor (long leg to GPIO, short leg to GND).
4. Ensure wiring matches the schematic:
   - **ESP32 Pin 2 → Resistor → LED Anode (+)**
   - **LED Cathode (-) → GND**

Step 2: Configure Firebase Realtime Database

1. Go to Firebase Console ↗.
2. Create a new project (if not already created).
3. Go to **Realtime Database → Create Database**.
4. Set rules to **public for testing** (students can change later for security):

```
JSON

{
  "rules": {
    ".read": true,
    ".write": true
  }
}
```

5. Copy the database URL (it looks like `https://your-project-id.firebaseio.com/led.json`).
6. Replace the URL in **app.js** and **ESP32 sketch** with your database URL.

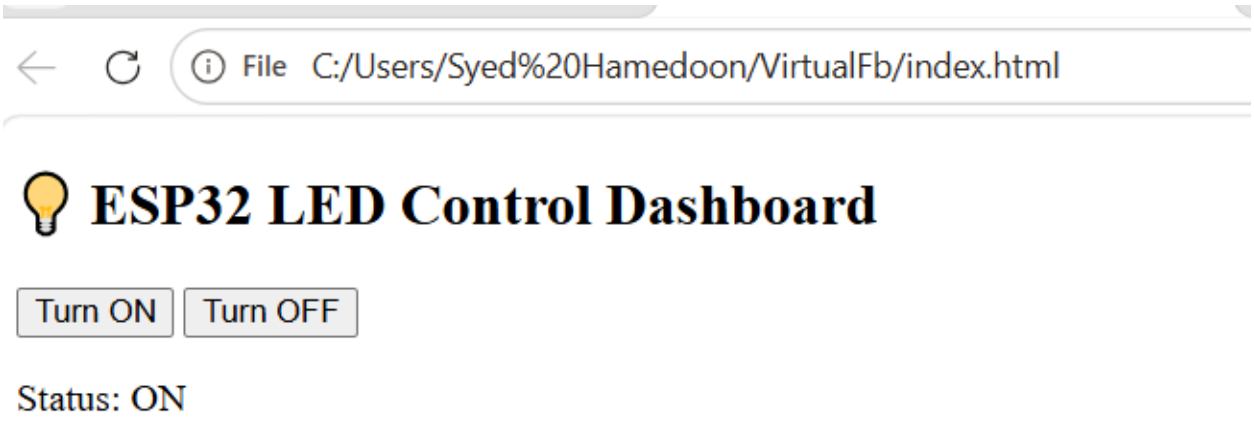


Step 3: Prepare the Web Dashboard

1. Create a folder on your computer called `ESP32_LED_Dashboard`.
2. Inside, create two files:
   - `index.html`
   - `app.js`
3. Paste the provided code:

index.html

My web-based dash board



index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>ESP32 LED Control</title>
</head>
<body>

  <h2>💡 ESP32 LED Control Dashboard</h2>

  <button onclick="turnOn()">Turn ON</button>
  <button onclick="turnOff()">Turn OFF</button>

  <p>Status: <span id="status">Unknown</span></p>

  <!-- Link JS file -->
  <script src="app.js"></script>

</body>
```

```
</html>
```

app.js

```
const baseURL = "https://virtualmcu-2ee29-default-rtdb.firebaseio.com";

function updateLED(state) {
  fetch(`${baseURL}/led.json`, {
    method: "PUT",
    headers: {
      "Content-Type": "application/json"
    },
    body: JSON.stringify({
      state: state,
      updatedAt: new Date().toLocaleString()
    })
  })
  .then(res => res.json())
  .then(data => {
    console.log("Updated:", data);
    document.getElementById("status").innerText = state === 1 ? "ON" : "OFF";
  })
  .catch(err => console.error(err));
}

function turnOn() {
  updateLED(1);
}

function turnOff() {
  updateLED(0);
}
```

Step 4: Program ESP32

1. Open Arduino IDE or Wokwi code editor.
2. Paste the provided `Sketch.ino` code:

Sketch.ino

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <ArduinoJson.h>
```

```
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// 🔥 IMPORTANT: point to specific key
String firebaseURL = "https://virtualmcu-2ee29-default-rtdb.firebaseio.com/led.json";

int ledPin = 2;

void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connecting...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("Connected!");
}

void loop() {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;

    http.begin(firebaseURL);
    int httpResponseCode = http.GET();

    if (httpResponseCode > 0) {
      String payload = http.getString();

      Serial.print("Firebase Data: ");
      Serial.println(payload);

      // ✅ Parse JSON properly
      StaticJsonDocument<200> doc;
      DeserializationError error = deserializeJson(doc, payload);

      if (!error) {
        int state = doc["state"];   // 🔥 get value

        Serial.print("Parsed State: ");
        Serial.println(state);
```

```
      if (state == 1) {
        digitalWrite(ledPin, HIGH);
        Serial.println("LED ON");
      } else {
        digitalWrite(ledPin, LOW);
        Serial.println("LED OFF");
      }

    } else {
      Serial.print("JSON Error: ");
      Serial.println(error.c_str());
    }

  } else {
    Serial.println("Error in HTTP request");
  }

  http.end();
 }

 delay(1000);
}
```

🔧 ✅ Step 1: Add Library in Wokwi

In Wokwi:

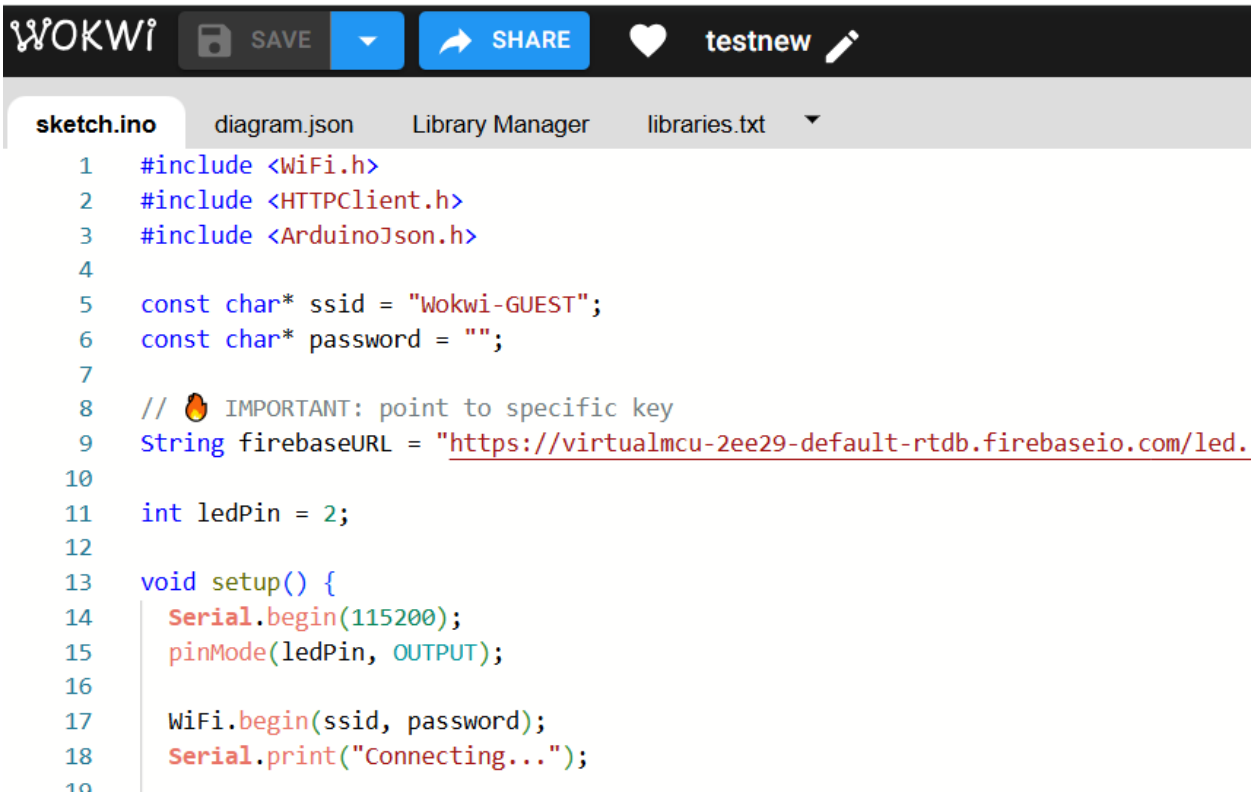
👉 Add to `libraries.txt`:

```
ArduinoJson
```

💻 ✅ Step 2: Replace Your Loop Code

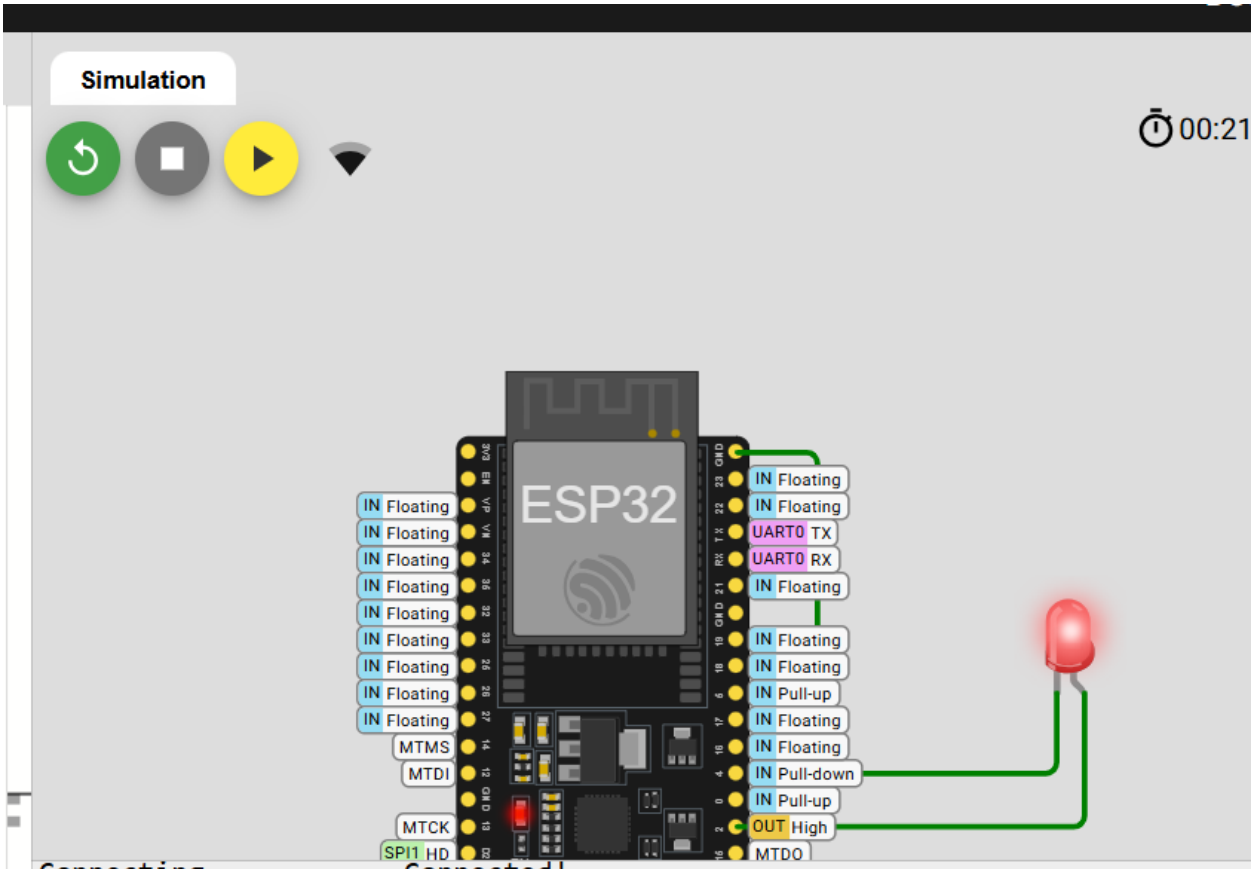
```
‹/› C++

#include <ArduinoJson.h>
```

WOKWI window



My WOKWI simulation Window

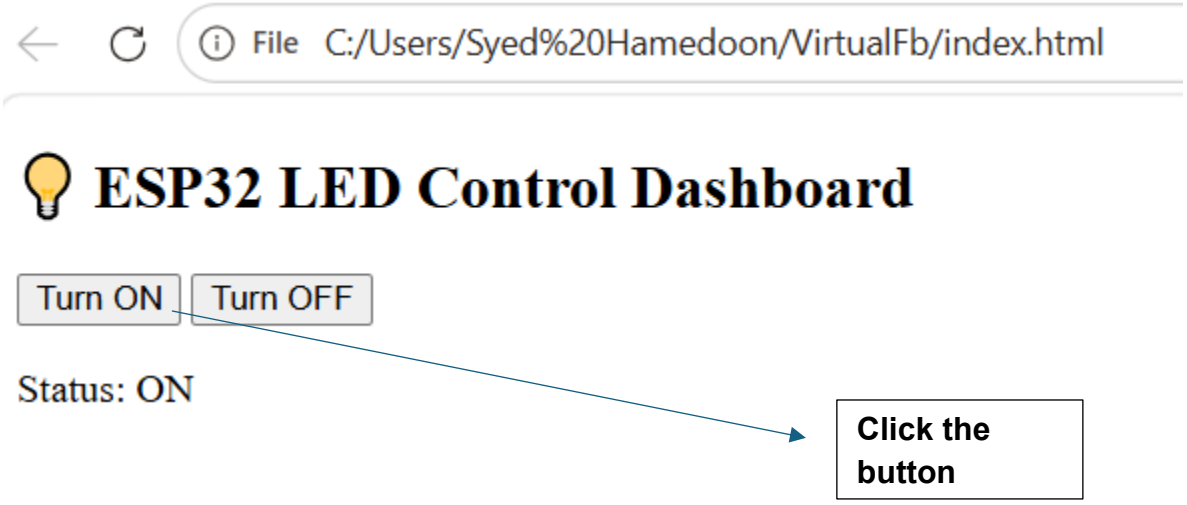
Connect LED anode with pin2 of EPS32 and cathode of LED with Ground and make circuit diagram accordingly. After save the code press the run button it take some times to connect then after connecting go to web based dash board.



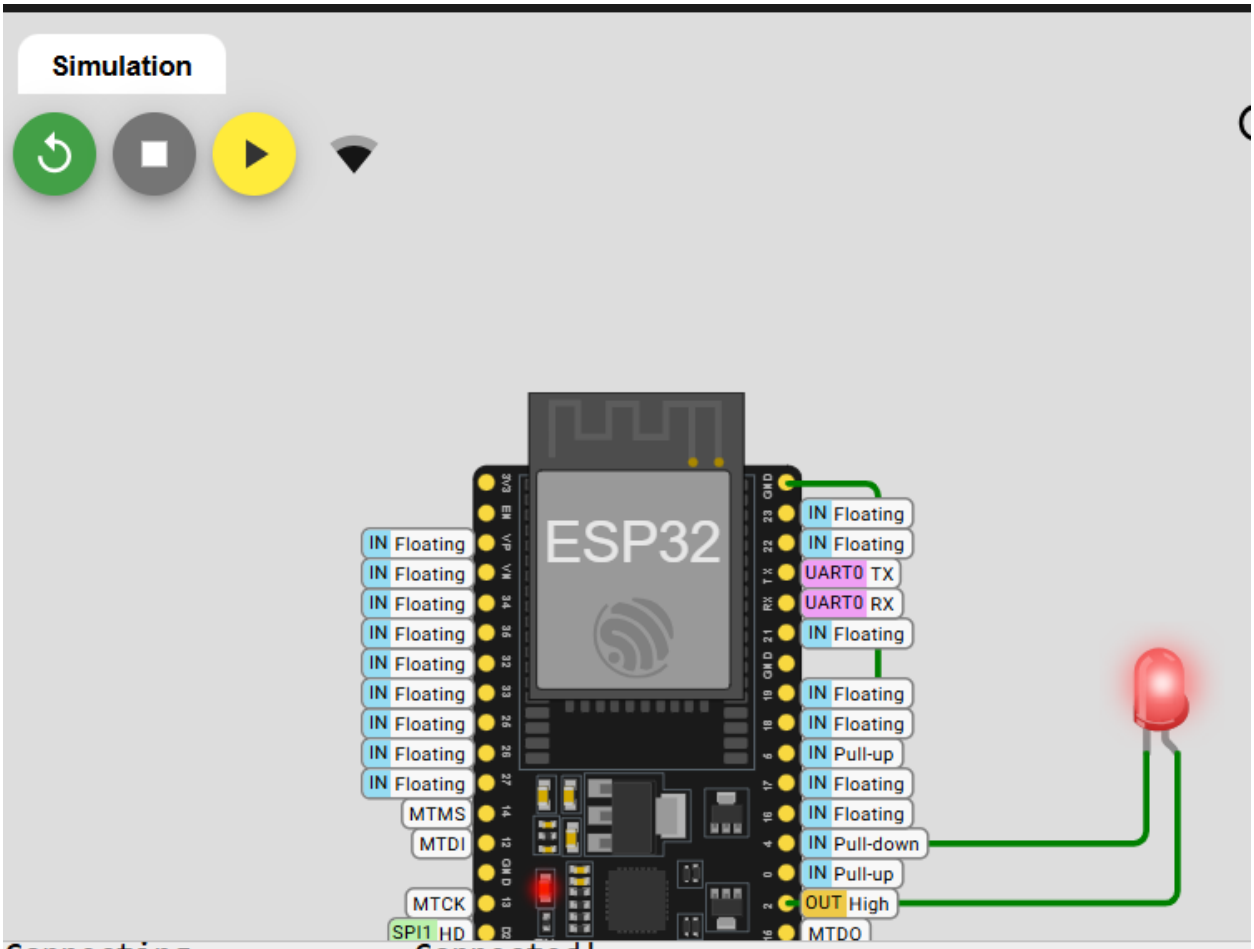

```
Connecting.....Connected!
Firebase Data: {"state":1,"updatedAt":"4/4/2026, 12:39:35 PM"}
Parsed State: 1
LED ON
Firebase Data: {"state":1,"updatedAt":"4/4/2026, 12:39:35 PM"}
Parsed State: 1
LED ON
```

Step 5: Test the Web Dashboard

1. Click **Turn ON** button → LED in simulation should turn on.
2. Click **Turn OFF** button → LED should turn off.
3. Observe **status text** on the dashboard updates to reflect the LED state.
4. Check **Serial Monitor** → it will print Firebase data, parsed state, and LED actions.



Observe the output on wokwi simulation diagram

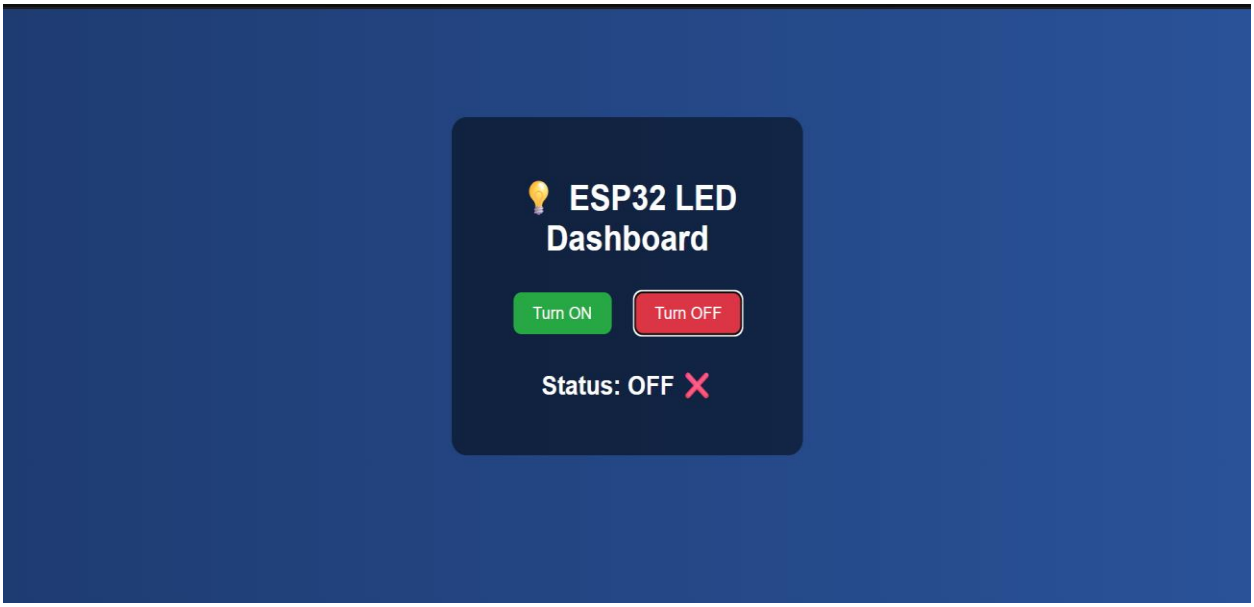


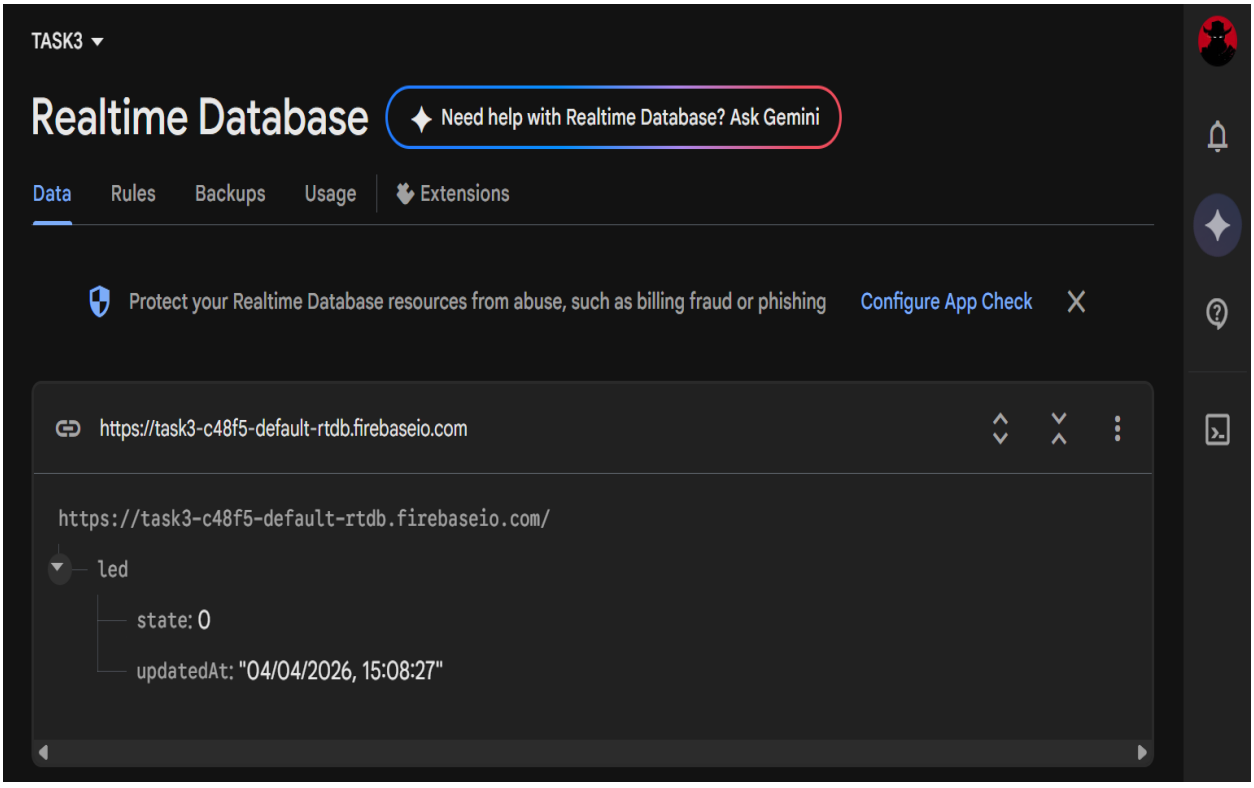

```
Connecting.....Connected!
Firebase Data: {"state":1,"updatedAt":"4/4/2026, 12:39:35 PM"}
Parsed State: 1
LED ON
Firebase Data: {"state":1,"updatedAt":"4/4/2026, 12:39:35 PM"}
Parsed State: 1
LED ON
```

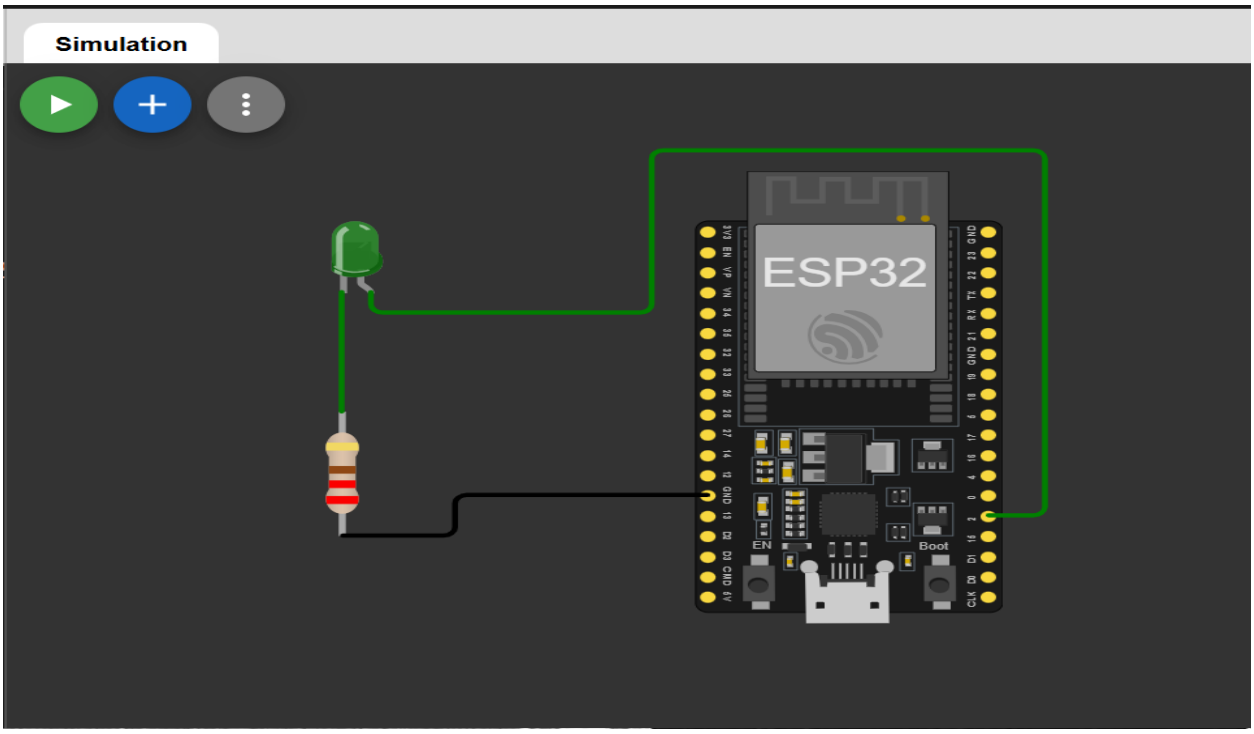
🧪 ✅ Expected Serial Output

```
Firebase Data: {"state":1,"updatedAt":"..."}
Parsed State: 1
LED ON
```

SOL:

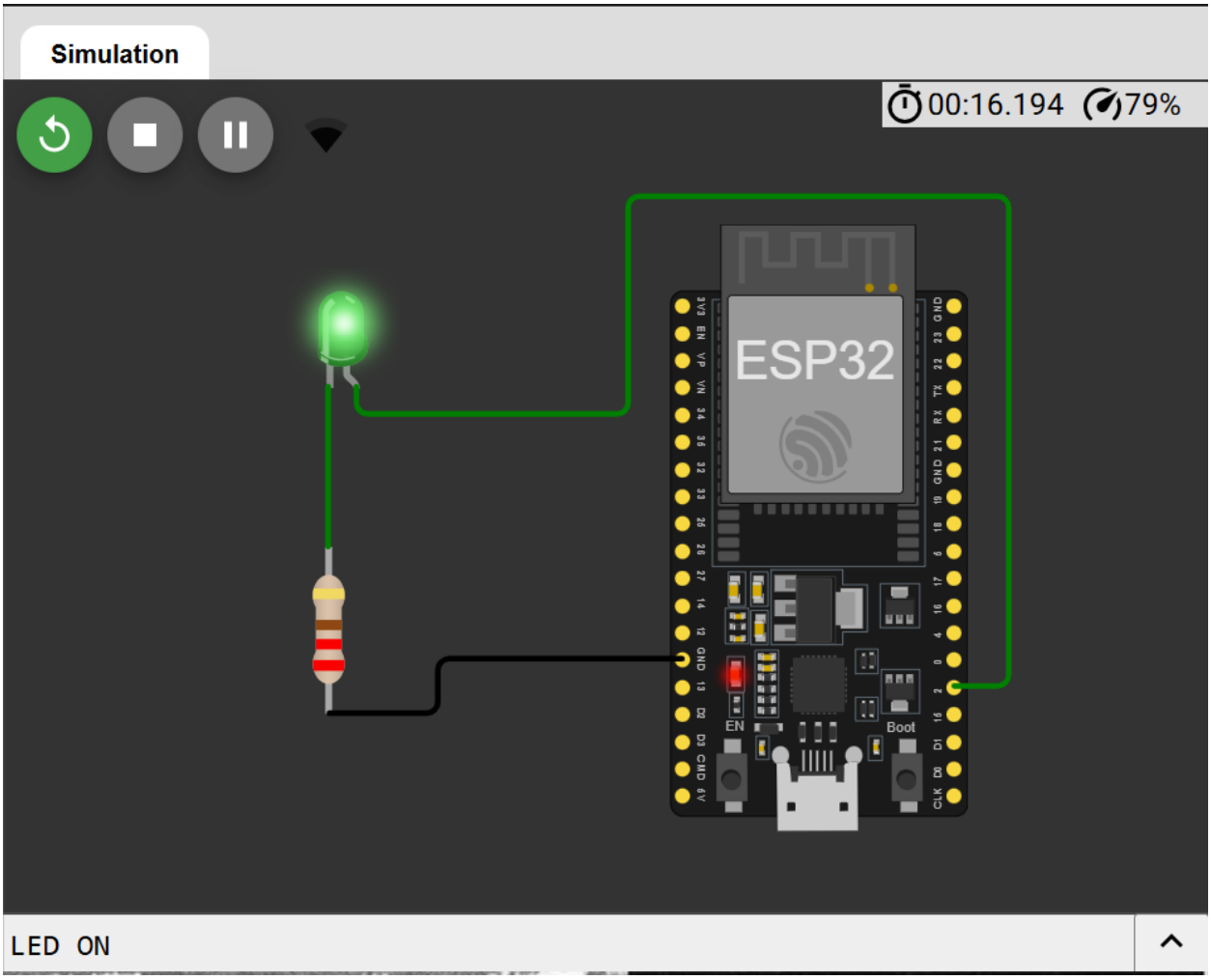

ESP32 LED Dashboard
Turn ON
Turn OFF
Status: OFF


TASK3
Realtime Database
Need help with Realtime Database? Ask Gemini
Data
Rules
Backups
Usage
Extensions
Protect your Realtime Database resources from abuse, such as billing fraud or phishing
Configure App Check
https://task3-c48f5-default-rtdb.firebaseio.com
https://task3-c48f5-default-rtdb.firebaseio.com/
led
state: 0
updatedAt: "04/04/2026, 15:08:27"




```
#include <WiFi.h>
#include <HTTPClient.h>
#include <ArduinoJson.h>

const char* ssid = "Wokwi-GUEST";
const char* password = "";

String firebaseURL = "https://task3-c48f5-default-rtdb.firebaseio.com/le

int ledPin = 2;

void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connecting...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected!");
}
```

```
27  void loop() {
28    if (WiFi.status() == WL_CONNECTED) {
29      HTTPClient http;
30
31      http.begin(firebaseURL);
32      int httpResponseCode = http.GET();
33
34      if (httpResponseCode > 0) {
35        String payload = http.getString();
36
37        StaticJsonDocument<200> doc;
38        DeserializationError error = deserializeJson(doc, payload);
39
40        if (!error) {
41          int state = doc["state"];
42
43          if (state == 1) {
44            digitalWrite(ledPin, HIGH);
45            Serial.println("LED ON");
46          } else {
47            digitalWrite(ledPin, LOW);
48            Serial.println("LED OFF");
49          }
50        }
51      }
```




```
entry 0x400805dc
Connecting.........
Connected!
LED ON
LED ON
LED ON
LED ON
```

https://task3-c48f5-default-rtdb.firebaseio.com

https://task3-c48f5-default-rtdb.firebaseio.com/

```
led
  state: 1
  updatedAt: "04/04/2026, 15:23:35"
```